Software Requirements Specification (SRS)

Project Name- Gesture Language Translator

1.Project Selection and Team Formation

1.1. Project selection

The project that I selected is Gesture Language Translator which is designed especially designed for people that use sign language as their primary language. This system identifies the hand gestures used in sign language and integrates its meaning to a suitable language.

This project is selected because:

- There are around 70 million people around the world that use sign language as their primary language.
- Most hearing individuals lack the ability to integrate the sign language.
- The project is so that it can break the barrier between the hearing and the deaf community.
- It enables people to integrate the sign language easily.

1.2. Team composition and roles

Team Member	Role	Responsibilities
Sandireddy Pramod Chowdary	Only Person in the Team	Planning, scheduling, coordination, instructor liaison, Architecture design, implementation, database management, Test case design, validation, defect tracking, SRS preparation, formatting, documentation control.

2.Team agreement

Researching about the previous or existing Asl models on how they work and the limitations faced by the systems and try to work on them together with the project everyday

3.Problem Statement Development

3.1. Problem Statement

There are 70 million people around the world that use sign language around the globe as a primary language despite this there is a silent barrier between the deaf community and the hearing world. Most hearing individuals lack the proficiency in sign language which leads to systematic exclusion in critical environments like healthcare, education or any critical situations. While speech to text technology is well established sign to text is still underdeveloped. Bridging this gap by developing aa accessible ai driven software that uses standard camera hardware that can translate complex hand gestures and facial patterns into real time text is essential to fostering global inclusivity, ensuring privacy, and providing equal chances for the deaf and mute community to fully participate in every aspect of modem society.

3.2. Affected Users

- **Deaf and Mute community.**
- **Specially Abled Schools.**
- **Hospitals.**
- **Corporate offices.**

3.3. proposed Solution

We propose developing an AI-based sign language translation system that uses a standard camera (mobile or laptop) to capture hand gestures and facial expressions. Using computer vision and deep learning, the system will convert sign language into real-time text, with optional text-to-speech support for two-way communication.

This accessible web/mobile application will help bridge the communication gap, promote inclusivity, and enable equal participation for the deaf and mute community in everyday and critical situations.

4. Stakeholder Analysis

4.1 Primary Stakeholders

1. **Deaf and Mute Users**

- Direct users of the system
- Use the system to translate sign language into text/speech

2. **Hearing Individuals**

- Use translated output to communicate with deaf users

4.2 Secondary Stakeholders

- Family members and caregivers
- Teachers and educational institutions
- Healthcare professionals
- Employers and workplace coordinators

4.3 Tertiary Stakeholders

- Development team
- Project guide and lab instructors
- Future developers and system maintainers

5. Stakeholder Interests and Influence

Stakeholder	Primary Interest	Influence
Deaf & Mute Users	Accurate real-time translation	High
Hearing Users	Easy communication	Medium
Educators / Doctors	Inclusive interaction	Medium
Project Guide	Academic outcomes	High

6. Stakeholder Register

ID	Stakeholder	Category	Interest	Influence
STK-01	Deaf & Mute Users	Primary	High	High
STK-02	Hearing Users	Primary	High	Medium
STK-03	Educators/Doctors	Secondary	Medium	Medium

STK-04	Project Guide	Evaluator	High	High

7. Scope Definition – In Scope

The following features are included within the scope of the system:

- User-friendly web/mobile interface
- Live video input using standard camera
- Hand gesture and facial expression detection
- Real-time sign-to-text translation
- Display of translated text output
- Secure and privacy-preserving procession.

8. Scope Definition – Out of Scope

The following features are excluded from the current scope:

- Hardware-based sensors or gloves
- Offline translation without camera access
- Multi-language sign language support (future enhancement)
- Speech-to-sign translation
- Cloud-based data storage

9. Scope Boundaries Document

The system boundaries are defined as follows:

- Single-user interaction at a time
- Real-time camera-based input only
- No external hardware dependency
- Standalone web or mobile execution

10. SRS Introduction

10.1 Purpose

This document specifies the functional and non-functional requirements for the AI-based Sign Language Translation System.

10.2 Scope

The system enables real-time translation of sign language gestures into readable text to improve communication accessibility.

10.3 Definitions and Acronyms

- **AI**: Artificial Intelligence
- **SRS**: Software Requirements Specification

- **CV**: Computer Vision

11. Overall Description

11.1 Product Perspective

The system is a standalone AI-powered accessibility application.

11.2 Product Functions

- Capture live video input
- Detect hand gestures and facial expressions
- Translate sign language into text
- Display translation in real time

11.3 User Characteristics

Users may have basic smartphone or computer usage skills.

11.4 Constraints

- Academic semester timeline
- Limited training dataset
- No paid APIs or hardware

12. Functional Requirements

FR-01: Video Input Capture

The system shall capture live video through a camera.
Acceptance Criteria: Video feed is displayed without lag.

FR-02: Gesture Recognition

The system shall identify sign language gestures from video input.
Acceptance Criteria: Gestures are correctly detected.

FR-03: Text Translation

The system shall convert recognized gestures into text.
Acceptance Criteria: Accurate text output is displayed in real time.

13. Non-Functional Requirements

- **NFR-01: Usability** – Easy to use for all users
- **NFR-02: Performance** – Translation within 2 seconds
- **NFR-03: Reliability** – Continuous operation without crashes
- **NFR-04: Maintainability** – Modular and well-documented code

14. Use Case Diagrams

Use case diagrams include:

- Capture Video

- Translate Sign Language
- View Translated Text

15. Glossary

Term	Definition
Sign Language	Visual language using hand gestures
Gesture	Hand or facial movement
Translation	Conversion to readable text

16. Project Charter

The project charter defines objectives, stakeholders, scope, and constraints for the Sign Language Translation System.

17. Stakeholder Register

A formal stakeholder register is maintained separately.

18. Complete SRS Document

This document constitutes the complete Software Requirements Specification.

19. Requirements Sign-Off

Name	Role	Signature	Date